# Directed Reading Program: Knot Theory

Abel Lynerd

University of Virginia

April 22nd, 2026

# Introduction

This semester, I worked with **Adam Friedman-Brown**, a Ph.D. student studying geometry, in UVA's Directed Reading Program. We decided to read **The Knot Book, by Colin Adams**, and explored several topics in knot theory and topology.

Today, I will be giving a rudimentary introduction to knot theory, focusing on the properties of **knot invariants** and culminating in a demonstration of the **Jones Polynomial**. I have crafted several sample knots out of paracord for you to use as a tactile aid. Questions are welcome at any time during the presentation.

# What is a knot?

Imagine taking a piece of string, tying it, and fusing the ends together. You now have a knot!

**In simplest terms:**

A mathematical **knot** is a closed curve in space that doesn't intersect itself anywhere.

A more rigorous topological definition is given as follows:

**Formal definition:**

A **knot** is an embedding of the circle $\mathbb{S}^1$ into three-dimensional Euclidean space $\mathbb{R}^3$.

# What is a projection?

There are **infinitely many** knots. Some of them are the same, and some are quite different.

**Projections:**

Take a knot $K$. A **projection** $K_p$ of $K$ is some knot that can be **transformed** into $K$ following rules we will soon establish.

Flatten your sample knot on your desk. That is one projection of the knot. Now pick the knot back up and manipulate it a little. When you flatten it on the table again, it's still the same knot– but it looks different, because it is a **different projection**.

# Rules about projections

It is necessary to establish some ground rules about what we can and cannot do to a knot when finding a new projection.

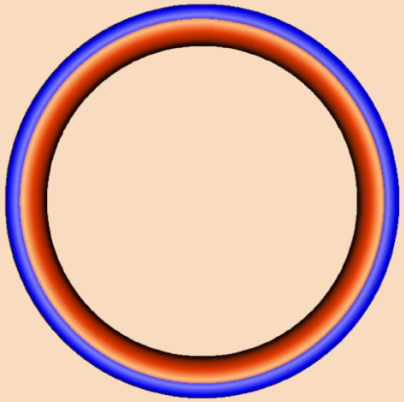
**You CAN:**

- Stretch and squeeze (this is called **ambient isotopy**)
- Twist and pull (these are defined more formally as **Reidemeister moves**)
- Generally, anything you can do with a physical loop of string

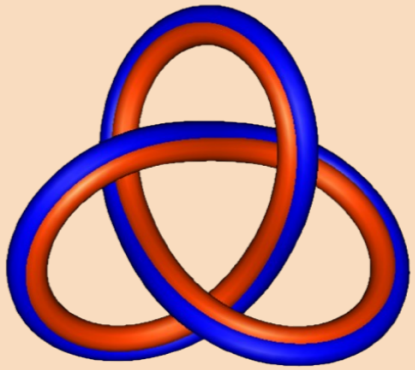
**You CANNOT:**

- Cut the string, manipulate the knot, and fuse it back together
- Pull a part of the knot infinitely tight to make it vanish

# Some example knots



Unknot



Trefoil ($3_1$)

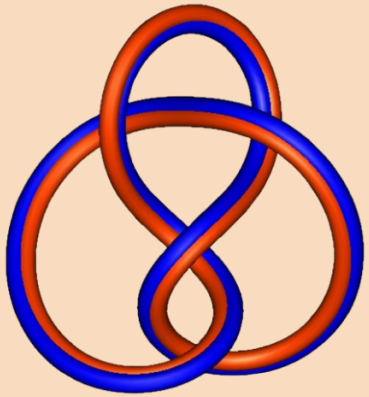
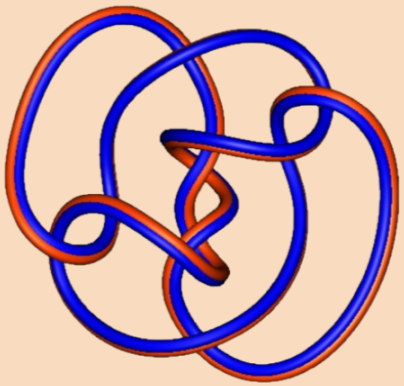
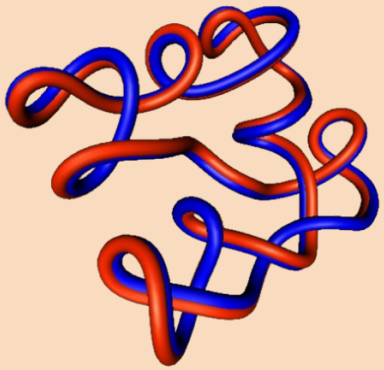


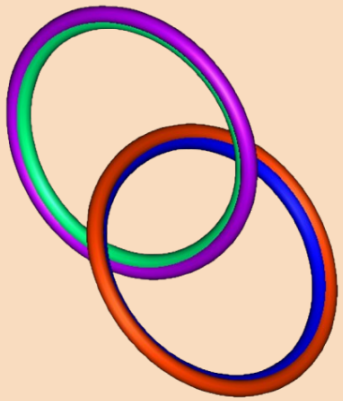
Figure-Eight ($4_1$)



$10_{144}$



Also the Unknot



Hopf Link

# Invariants

So, you may be asking yourself: How *exactly* can one rigorously tell two knots apart? If so, you're not alone– this question is one of the foundational problems of knot theory, and topologists are still finding new ways to prove different knots are distinct.

## Invariants

An **invariant** is a property that is constant between all projections of a knot.

Note that two distinct knots can have the same properties under certain invariants. For this reason, it is relatively easy to prove two knots are distinct– simply find an invariant where they differ– but much harder to prove that two knots are the same.

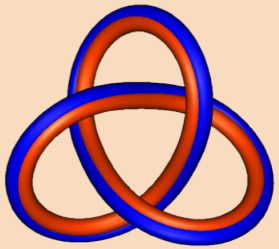
# Unknotting Number

One of the simplest invariants, for example, is the **unknotting number**. This is the number of **crossings** needed to be flipped to turn the knot into a projection of the unknot.

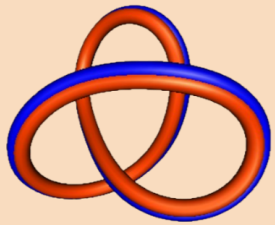
## Crossing

In a knot projection, a **crossing** is where one part of the knot passes over another. A **crossing flip** passes the knot through itself to make the overstrand the understrand, and vice versa.

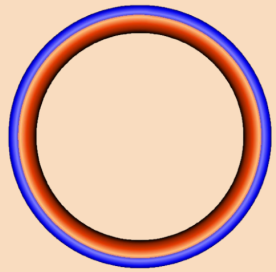
We can see that the unknotting number of the trefoil is 1:



Take the trefoil,



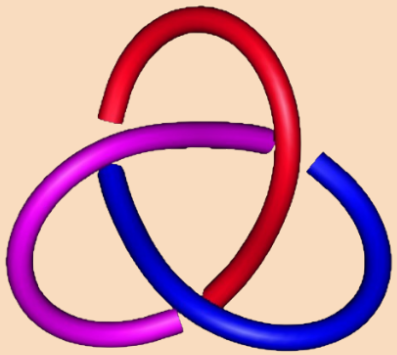
flip a crossing,



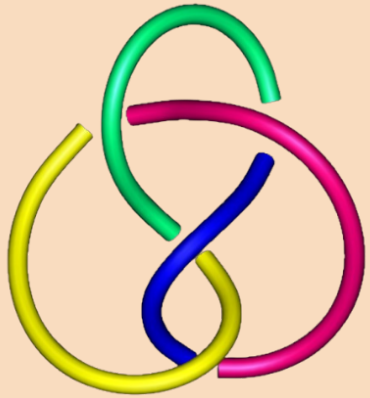
and get the unknot.

# Tricolorability

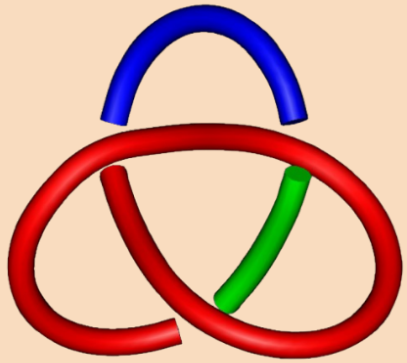
Another invariant we can calculate is **tricolorability**. For a knot to be tricolorable, any of its projections can be split at the under-crossings and have each strand colored one of three colors (using at least two) such that at every crossing, either all three colors meet up or just one.



We can see that the trefoil is tricolorable...



...but the figure-eight is not...



... and neither is this projection of the unknot...

Hence, we have that the trefoil and the figure-eight must be distinct, but we do **not** have that the figure-eight and the unknot are the same.

# The Bracket Polynomial

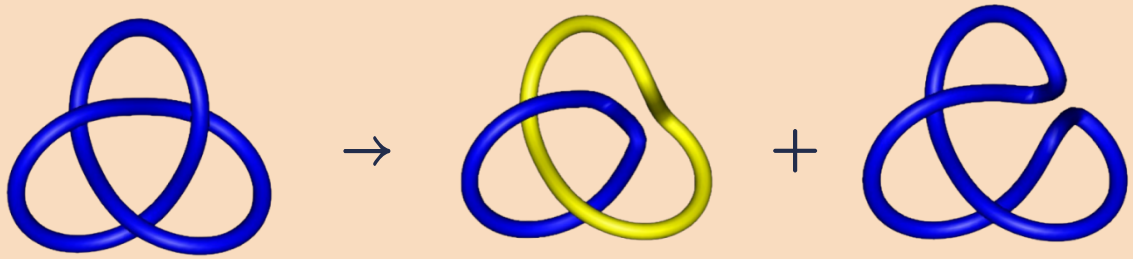
Knots can also be formulated into polynomials. The simplest of these is called the **bracket polynomial**, which we can calculate by splitting up each crossing (called a **Skein relation**) under the following rules:

- The unknot: $\langle \bigcirc \rangle = 1$
- Positive crossings: $\langle \times \rangle = A\langle )( \rangle + A^{-1}\langle \asymp \rangle$
- Negative crossings: $\langle \times \rangle = A\langle \asymp \rangle + A^{-1}\langle )( \rangle$
- Unions with the unknot: $\langle L \cup \bigcirc \rangle = (-A^2 - A^{-2}) \langle L \rangle$

We split every crossing like this until every term is reduced to an unknot, leaving the polynomial entirely in terms of the variable $A$.

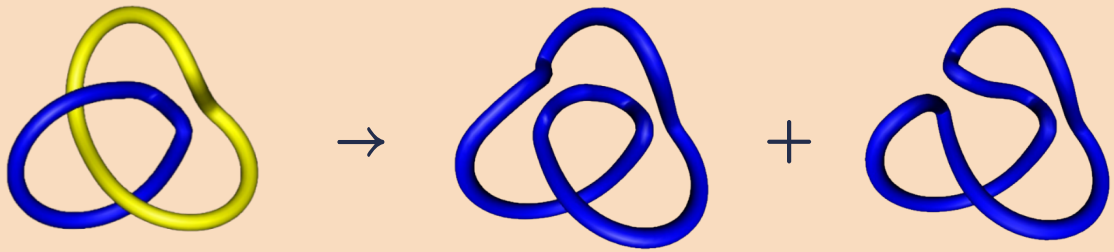
# Bracket Polynomial of the Trefoil

Here we will calculate the bracket polynomial of the trefoil. We can start by splitting the top right crossing of this projection:



Trefoil $\quad A \quad A^{-1}$

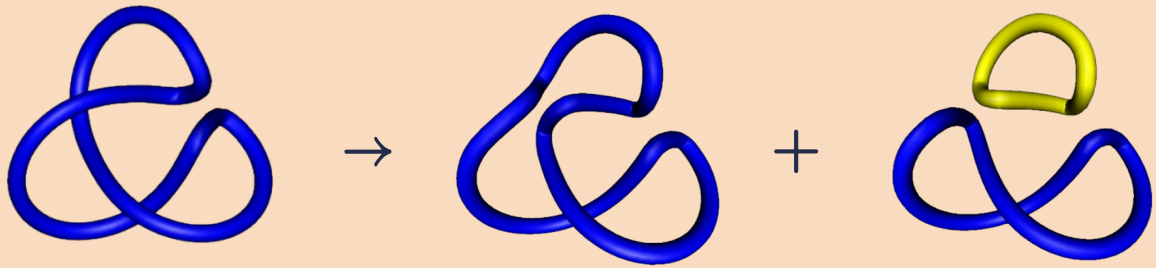
Next we will split the first term (which is equal to the Hopf Link):



$A \quad A(A) = A^2 \quad A(A^{-1}) = 1$

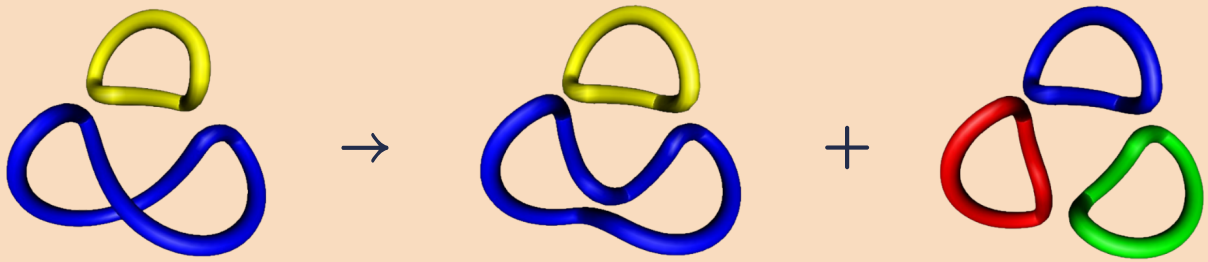
# Bracket Polynomial of the Trefoil (continued)

We will now briefly double back and split the second term of the first part:



$A^{-1}$ $\qquad$ $A^{-1}(A) = 1$ $\qquad$ $A^{-1}(A^{-1}) = A^{-2}$

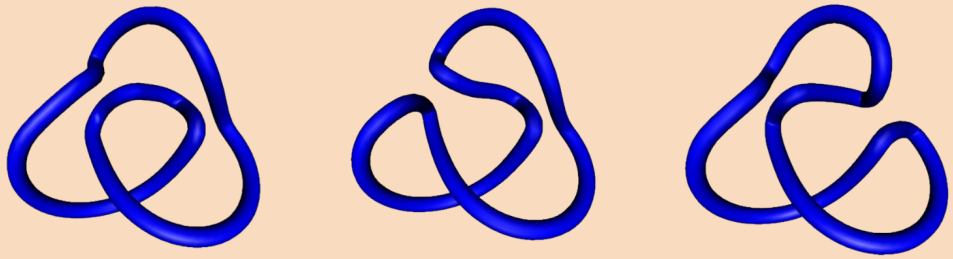
And then the second term of that:



$A^{-2}$ $\qquad$ $A^{-2}(A) = A^{-1}$ $\qquad$ $A^{-2}(A^{-1}) = A^{-3}$

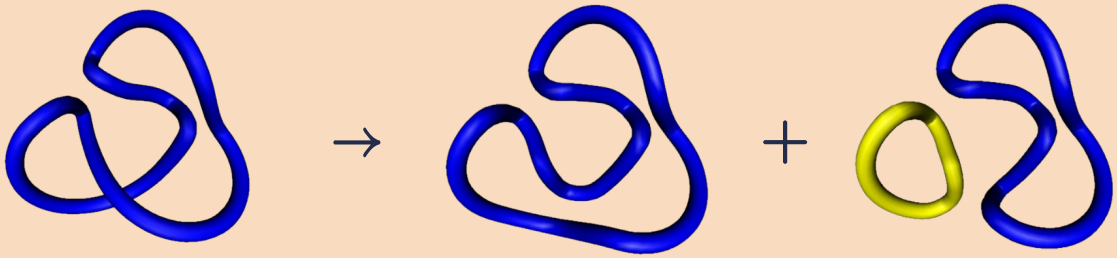
# Bracket Polynomial of the Trefoil (continued)

Consider the remaining one-crossing terms:



(a) $A^2$ (b) 1 (c) 1

We can see that (b) can be stretched and squeezed into (c), so their split can be calculated in the same way:

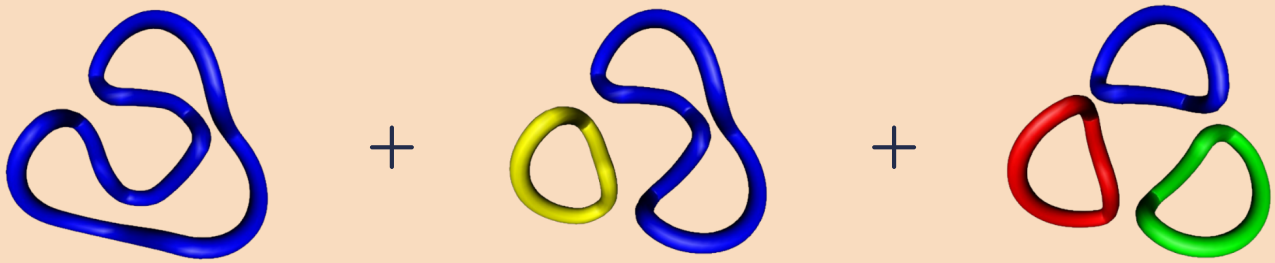


1 $1(A) = A$ $1(A^{-1}) = A^{-1}$

The split of (a) can be calculated in the same way but with the coefficients swapped.

# Bracket Polynomial of the Trefoil (continued)

We have now fully reduced the trefoil into a sum of knots with no crossings:



$3A$ $\qquad$ $3A^{-1} + A^3$ $\qquad$ $A^{-3}$

and can now apply the union rule to see that the bracket polynomial of the trefoil is:

$$3A + (-A^2 - A^{-2})(3A^{-1} + A^3) + (-A^2 - A^{-2})^2 A^{-3}$$
$$= 3A - 3A - A - 3A^{-3} - A^5 + A + 2A^{-3} + A^{-7}$$
$$= -A^5 - A^{-3} + A^{-7}$$

# The Auxiliary Polynomial

You may have noticed that a bracket polynomial changes when a part of the projection is "twisted" meaning it is not actually an invariant.

We can refine the bracket polynomial of a knot $K$ into something called the **auxiliary polynomial** by multiplying it by $(-A^3)^{-w(K)}$, where $w(K)$ represents the **writhe** of $K$.

**Writhe**

The **writhe** of a curve $C$ is a measure of its "coiliness;" for a knot projection it can be determined by **orienting** the knot and subtracting the number of negative crossings from the number of positive ones.

This negates the effect of a twist, making the auxiliary polynomial a true invariant.

# The Jones Polynomial

In 1984, Vaughan Jones, an operator algebraist from New Zealand, found that the Auxiliary polynomial could be used to create a powerful invariant. The **Jones polynomial**, which is created by replacing $A$ with $t^{-\frac{1}{4}}$, is useful because over certain subsets of the knots it is **unique**.

## Uniqueness

An invariant is **unique** if two knots that have the same properties under it must be the same. In other words, there is no more than one knot to match each polynomial.

Although it is quite useful for distinguishing most knots, there are still certain special cases where two knots have the same Jones Polynomial.

# The Jones Polynomial of the Trefoil

We can first find the auxiliary polynomial of the trefoil, calculating its writhe to be 3:

$$(-A^3)^{-3}(-A^5 - A^{-3} + A^{-7})$$
$$-A^{-9}(-A^5 - A^{-3} + A^{-7})$$
$$= A^{-4} + A^{-12} - A^{-16}$$

Then, we can plug in $A = t^{-\frac{1}{4}}$ to get its Jones Polynomial:

$$\left(t^{-\frac{1}{4}}\right)^{-4} + \left(t^{-\frac{1}{4}}\right)^{-12} - \left(t^{-\frac{1}{4}}\right)^{-16} = t + t^3 + t^4.$$

# Thank you all for attending my presentation!

Special thanks to my mentor Adam, and to all of the other grad students and faculty who generously spent their time at the blackboard with me, working through these new concepts. I would also like to thank Alan Hale and Sean Dermody for helping me put together the souvenir knots.

Knot images were generated using Dr. Rob Scharein's KnotPlot software. Bracket polynomial notation was written with TikZ.